

A field-level dark-siren hierarchy: mathematical specification

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object	source of truth (master @ 0c5b3db)
$dN_{\text{miss}} = (1 - C) dN_{\text{exp}} Q$	redshift/completion.py:1280
low-rank operator $\Phi = K_{XZ} L^{-\top}$	redshift/lognormal_completion.py:635--655
MAP objective $J(\xi)$	redshift/lognormal_completion.py:660--700
Laplace ensemble ξ_m	redshift/lognormal_completion.py:868--887
log Q evaluation (both modes)	redshift/lognormal_completion.py:889--978
per- z mean-one renormalisation	redshift/lognormal_completion.py:502--560
log $\mathcal{L} = \text{LSE}_m \log \mathcal{L}_m - \log M$	likelihood/core.py:1274
two-node member gather (the seam)	redshift/prior.py:711--758
$K \geq 2$ shared-latent stacked solve	cli/build_joint_lognormal_completion.py:239--290

Table 1: Anchors for the current-state equations of §2.

1 Notation and the objects that already exist

Sky pixels are indexed $p \in \{1, \dots, N_{\text{pix}}\}$ in HEALPix RING ordering at $N_{\text{side}} = 64$ ($N_{\text{pix}} = 49\,152$, $\Omega_{\text{pix}} = 4\pi/N_{\text{pix}}$). Redshift is carried on the package grid $\{z_n\}_{n=1}^{N_{\text{grid}}}$ ($N_{\text{grid}} = 1086$). A *voxel* is $v = (p, z)$. Coarse radial shells used by the field fit are indexed $g \in \{1, \dots, G_s\}$ with edges $\{z_g^-, z_g^+\}$. Catalogs (tracers) are indexed $k \in \{1, \dots, K\}$.

Cosmological and population parameters are Λ (containing H_0); survey parameters are $\theta = (\log_{10} n_0, \delta, \theta_{\text{sel}}, b_{\text{miss}}, \dots)$, per catalog where suffixed. Throughout, the *expected all-galaxy comoving density* is

$$\bar{n}(z; \Lambda, \theta) \equiv \frac{dN_{\text{exp}}}{dz d\Omega} = n_0 (1+z)^\delta \frac{dV_c}{dz d\Omega}(z; \Lambda), \quad (1)$$

which is the code object `grids.dN_exp`. $C(z; \theta) \in [0, 1]$ is the completeness curve; under `c_mode="per_pixel"` it is $C_p(z)$ (a per-pixel KDE ratio), under `"aggregate"` it is the sky-aggregate $\bar{C}(z)$, and under `"selection"` it is the parametric $C_{\text{sel}}(z; \theta_{\text{sel}})$ formed from a truncated luminosity function (optionally stratified, giving a per-pixel *depth* routing $C_{s(p)}(z; \theta_{\text{sel}})$ through a stamped stratum map — single-catalog only). The production 259-event run uses `c_mode=selection`, no stratification, and *no LSS at all*: it runs `--use_lss false` with `lss_completion none` (`logs/ns_joint_sel_1119811.out:68, 73, 2`) so the field-level path would be the one that puts LSS into a production run of this pipeline; no production run on this line has ever carried it.

2 The current state as one generative model

The pieces are usually described as three separate machines (a builder, a table, a likelihood). They are one generative model, and writing it as one is what exposes where it is exact and where it is a plug-in.

2.1 The galaxy point process and its thinning

Conditional on a latent log-density contrast field $s(p, z)$, galaxies are an inhomogeneous Poisson process on (p, z) with intensity

$$\Lambda_{\text{gal}}(p, z) = \bar{n}(z; \Lambda, \theta) \exp\left[b_{\text{gal}} s(p, z) - \frac{1}{2} b_{\text{gal}}^2 \sigma^2(p, z)\right], \quad \sigma^2(p, z) \equiv \text{Var}[s(p, z)], \quad (2)$$

the lognormal density model. The shift $-\frac{1}{2} b^2 \sigma^2$ enforces $\mathbb{E}_{\text{prior}}[e^{bs - \frac{1}{2} b^2 \sigma^2}] = 1$, so (2) has prior-mean intensity exactly $\bar{n}$: the field *redistributes* galaxies and does not create them. This mean-one

convention is the single most load-bearing convention in the whole stack and reappears three times below.

The survey observes each galaxy independently with probability $f_p C(z; \theta)$, where $f_p \in [0, 1]$ is the per-pixel areal completeness ($f_p = 1 - \text{masked_frac}_p$; $f_p \equiv 1$ in today's code) and C is the magnitude/flux completeness. By the Poisson thinning theorem the observed (*catalogued*) and unobserved (*missing*) galaxies are two **independent** Poisson processes,

$$\Lambda_{\text{cat}}(p, z) = f_p C(z; \theta) \Lambda_{\text{gal}}(p, z), \quad (3)$$

$$\Lambda_{\text{miss}}(p, z) = [1 - f_p C(z; \theta)] \Lambda_{\text{gal}}(p, z), \quad (4)$$

and, crucially,

$$p(\text{missing galaxies} \mid \text{catalogued galaxies}, s) = p(\text{missing galaxies} \mid s). \quad (5)$$

Equation (5) is exact, and it is the entire resolution of the double-conditioning question of §3.4. It says: *once the field is fixed, the catalog carries no further information about what is missing*. Everything the catalog knows about the missing galaxies, it knows through s .

2.2 The dark-siren redshift–sky prior as it is coded

The likelihood does not integrate (3) — it uses the catalog itself. For a dark siren with parameters $(z, \hat{n}_p)$,

$$p(z, p \mid \Lambda, \theta, Q) = \frac{N_{\text{obs},p} \mathcal{K}_p(z) + [1 - C(z; \theta)] \bar{n}(z; \Lambda, \theta) Q_p(z)}{\mathcal{Z}(\Lambda, \theta, Q)}, \quad (6)$$

[code: `prior.py:668--673, :755--758`]

where $N_{\text{obs},p}$ is the integer galaxy count in pixel p and $\mathcal{K}_p(z)$ the pixel's photo- z kernel density (`eval_log_catalog_prior_state`). The normaliser $\mathcal{Z}$ is either per-pixel ($\mathcal{Z}_p$, “conditional” convention) or survey-global (“field” convention, `completion.py:1780--1810`). The consumed missing density is

$$\frac{dN_{\text{miss}}}{dz d\Omega}(p, z) = [1 - C(z; \theta)] \bar{n}(z; \Lambda, \theta) Q_p(z), \quad Q_p(z) \equiv e^{\text{clip}(\log Q_p(z), \pm 7)}, \quad (7)$$

relaxed to $dN_{\text{miss}} = dN_{\text{exp}}$ (i.e. $C := 0$, $Q := 1$) above the survey depth z_{depth} . [code: `completion.py:1280, 1295`]

Comparing (7) with (4) identifies

$$Q_p(z) \equiv \exp\left[b_{\text{miss}} s(p, z) - \frac{1}{2} b_{\text{miss}}^2 \sigma^2(p, z)\right] \quad \text{with } f_p \equiv 1. \quad (8)$$

So Q is the exponentiated latent field, mean-one normalised, and nothing else. Its only defect as written is that in the current code it is a **fixed table read from disk** rather than a function of the sampled parameters.

2.3 The offline field model: Poisson-lognormal on a low-rank basis

Let $\{Z_j\}_{j=1}^M$ be inducing nodes on $\mathbb{S}^2 \times \zeta$ with $\zeta = \log(1 + z)$ (a Fibonacci sphere of M_{sph} points crossed with M_z nodes uniform in ζ , $M = M_{\text{sph}} M_z$), and $k(\cdot, \cdot)$ the separable kernel $k = k_{\text{sph}} \otimes k_{\zeta}$ with amplitude a and lengthscales $(\ell_{\text{ang}}, \ell_{\zeta})$. With $LL^\top = K_{ZZ} + j\mathbb{K}$ define the Nyström design

$$\Phi = K_{XZ} L^{-\top} \in \mathbb{R}^{V \times M}, \quad s = \Phi \xi, \quad \xi \sim \mathcal{N}(0, \mathbb{K}_M), \quad (9)$$

so that $\text{Cov}(s) = \Phi \Phi^\top = K_{XZ}(K_{ZZ} + j\mathbb{K})^{-1}K_{ZX}$ and the per-voxel prior variance is $\sigma_{\text{prior}}^2(v) = \sum_i \Phi_{vi}^2$. Φ is never materialised: it factorises exactly as $\Phi = \Phi_{\text{sph}} \otimes \Phi_\zeta$ (a dense Φ at $M = 3780$ would be 47.6 GB).

The fit is over solve voxels $v = (p, g)$ carrying integer counts N_v and a base

$$\bar{\lambda}_v = \int_{z_g^-}^{z_g^+} C(z; \theta_{\text{ref}}) \bar{n}(z; \Lambda_{\text{ref}}, \theta_{\text{ref}}) dz, \quad (10)$$

which is p -independent under `c_mode` $\in \{\text{aggregate, selection}\}$ and p -dependent under `per_pixel`. The MAP problem is

$$J(\xi) = \frac{1}{2} \|\xi\|^2 + \sum_v [\lambda_v - N_v \log \lambda_v], \quad \lambda_v = \bar{\lambda}_v \exp \left[b(\Phi \xi)_v - \frac{1}{2} b^2 \sigma_{\text{prior}}^2(v) \right], \quad (11)$$

[code: `lognormal_completion.py:660--700`]

$$\nabla J = \xi + b \Phi^\top (\lambda - N), \quad H = \nabla^2 J = \mathbb{K} + b^2 \Phi^\top \text{diag}(\lambda) \Phi. \quad (12)$$

J is convex; $H \succeq \mathbb{K}$ always, so the problem is well posed even when $V < M$. The field is clipped at $|s| \leq 10$ with a KKT projection so that objective, gradient and Hessian describe the same clipped function.

The Laplace posterior is $p(\xi | \{N\}) \approx \mathcal{N}(\hat{\xi}, H^{-1})$, sampled as

$$\xi_m = \hat{\xi} + L_H^{-\top} g_m, \quad g_m \sim \mathcal{N}(0, \mathbb{K}_M), \quad L_H L_H^\top = H. \quad (13)$$

2.4 The two Q evaluations, and the variance shift

Let $\phi_u \in \mathbb{R}^M$ be the Nyström row at an *output* voxel $u = (p, z)$ on the fine grid, and

$$\sigma_{\text{prior}}^2(u) = \|\phi_u\|^2, \quad \sigma_{\text{post}}^2(u) = \phi_u^\top H^{-1} \phi_u = \|L_H^{-1} \phi_u\|^2. \quad (14)$$

Lemma 1 (posterior-mean Q and its variance shift). *Under the Laplace posterior (13),*

$$\log \mathbb{E}[Q_u] = b \phi_u^\top \hat{\xi} - \frac{1}{2} b^2 \left[\sigma_{\text{prior}}^2(u) - \sigma_{\text{post}}^2(u) \right]. \quad (15)$$

Proof. $Q_u = \exp[b \phi_u^\top \xi - \frac{1}{2} b^2 \sigma_{\text{prior}}^2(u)]$ by (8)–(9). For $\xi \sim \mathcal{N}(\hat{\xi}, H^{-1})$ and any a , $\mathbb{E}[e^{a^\top \xi}] = \exp(a^\top \hat{\xi} + \frac{1}{2} a^\top H^{-1} a)$. Put $a = b \phi_u$. $\square$

Equation (15) is the table the GW likelihood consumes today [code: `lognormal_completion.py:967`], and the per-draw member tables are

$$\log Q_u^{(m)} = b \phi_u^\top \xi_m - \frac{1}{2} b^2 \sigma_{\text{prior}}^2(u) \quad (16)$$

[code: `lognormal_completion.py:973`]

so that $\frac{1}{M} \sum_m Q_u^{(m)} \rightarrow \mathbb{E}[Q_u]$ exactly, by Lemma 1. Data-free voxels have $\sigma_{\text{post}}^2 = \sigma_{\text{prior}}^2$ and $\phi_u^\top \hat{\xi} \rightarrow 0$, so $Q \rightarrow 1$: “no information, no modulation.”

Remark 1 (the plug-in gap, stated exactly). The GW likelihood is a nonlinear functional of Q — Q appears inside log in (6) and inside $\log \mu$ in the selection term — so

$$\log \mathbb{E}_\xi [\mathcal{L}(Q(\xi))] \neq \log \mathcal{L}(\mathbb{E}_\xi [Q]). \quad (17)$$

Running the deterministic table is a *plug-in*, exact only to first order in the posterior spread of s . The member ensemble is the correction; its own error budget is Proposition 3.

2.5 The per- z mean-one budget constraint as a gauge condition

Proposition 1 (Q has no monopole freedom). *For any positive function $\alpha(z)$, the map*

$$\mathcal{G}_\alpha : \quad Q_p(z) \mapsto \alpha(z) Q_p(z), \quad [1 - C(z; \theta)] \bar{n}(z; \Lambda, \theta) \mapsto \frac{[1 - C(z; \theta)] \bar{n}(z; \Lambda, \theta)}{\alpha(z)} \quad (18)$$

leaves dN_{miss} in (7), and therefore the entire GW likelihood, exactly invariant. Hence the per- z monopole of Q is not identified by the GW data; it is identified only by whatever fixes $(n_0, \delta, \theta_{\text{sel}})$.

The code fixes this gauge, at build time, by the condition

$$\log Q_p(z) \longleftarrow \log Q_p(z) - \log \left[\frac{\sum_{p \in \mathcal{F}} w_p(z) Q_p(z)}{\sum_{p \in \mathcal{F}} w_p(z)} \right], \quad w_p(z) = [1 - C_p(z)] \bar{n}(z), \quad (19)$$

applied over the *fitted footprint* $\mathcal{F}$, to the MAP table and to **each member independently**. [code: `lognormal_completion.py:502--560; build_lognormal_completion.py:582, 990`]

Proposition 2 (when the gauge fixing is exactly what the likelihood consumes). *Let $\mathcal{B}(z) = \sum_{p=1}^{N_{\text{pix}}} [1 - C_p(z; \theta)] \bar{n}(z; \theta) Q_p(z)$ be the budget the likelihood actually integrates, and $\mathcal{B}_0(z)$ the same sum with $Q \equiv 1$. Then:*

1. *Under $\text{c_mode} \in \{\text{aggregate, selection}\}$ in the `gp3d` builder, $\mathcal{F}$ is the whole sky (`fit = np.arange(n_pix)`) and $w_p(z) = w(z)$ is p -independent (`np.tile`). Then (19) reduces to $\langle Q \rangle_{\text{sky}} = 1$, and*

$$\mathcal{B}(z) = [1 - C(z; \theta)] \bar{n}(z; \theta) \sum_p Q_p(z) = [1 - C(z; \theta)] \bar{n}(z; \theta) N_{\text{pix}} = \mathcal{B}_0(z) \quad (20)$$

exactly, for every z , every member, and every θ — the gauge condition is θ -free because the weights are.

2. *Under $\text{c_mode} = \text{"per_pixel"}$, $\mathcal{F}$ is the occupied subset and w_p depends on p and on θ . Then $\mathcal{B}(z) \neq \mathcal{B}_0(z)$ in general, by two separate mechanisms: (i) the off-footprint pixels ($Q \equiv 1$) are outside the constraint but inside the consumed sum, and (ii) the weights used at build time are evaluated at θ_{ref} .*

Remark 2. Proposition 2(1) is a stronger statement than the plan’s §4.2 assumes. In the *production* configuration (`c_mode=selection`) the mean-one identity is already exactly the consumed identity; the discrepancy the reviews flag is real, but confined to `per_pixel`. This matters for the proposed model because it says the budget/placement split survives a change of θ *without any frozen weight operator* whenever the completeness base is sky-uniform.

Remark 3 (the renormalisation does not commute with the ensemble mean). Write Π for the map (19). Π is nonlinear in Q , so

$$\frac{1}{M} \sum_m \Pi[Q^{(m)}] \neq \Pi[\mathbb{E}[Q]]. \quad (21)$$

The deterministic table (renormalise the posterior mean) and the member ensemble (renormalise each draw) therefore target slightly different objects. The bias is $O(\text{Var}_{\text{sky}}(Q)/N_{\text{pix}})$ per shell and is second order, but it is not zero, and it is the reason the members carry “placement uncertainty only, zero budget uncertainty” — a statement that is a modelling choice, not a theorem.

2.6 Member-ensemble marginalisation

With members $\{Q^{(m)}\}_{m=1}^{M_{\text{draw}}}$ the likelihood is

$$\log \mathcal{L}(\Lambda, \theta) = \log \frac{1}{M_{\text{draw}}} \sum_m \mathcal{L}_m(\Lambda, \theta) = \text{LSE}_m \log \mathcal{L}_m - \log M_{\text{draw}}, \quad (22)$$

[code: `likelihood/core.py:1274`]

$$\log \mathcal{L}_m = \sum_{i=1}^{N_{\text{ev}}} \log \int d\vartheta \frac{p(d_i|\vartheta) \pi_{\text{pop}}(\vartheta|\Lambda) p_z(z, p|\Lambda, \theta, Q^{(m)})}{\pi_{\text{PE}}(\vartheta)} - N_{\text{ev}} \log \mu(\Lambda, \theta, Q^{(m)}). \quad (23)$$

[code: `likelihood/core.py:1249--1262`]

The member index enters **both** the per-event (PE) term and the selection term μ ; `core.py:1372--1380` enforces this as an all-or-none guard, and the per-member N_{eff} /variance guard already runs inside the member `vmap`. This is the correct structure: a member is one realisation of the universe, and the same universe must be used to place hosts and to compute the detected fraction.

Proposition 3 (Jensen bias of (22)). *(22) is an unbiased estimator of $\mathcal{L}$ but a biased estimator of $\log \mathcal{L}$. If $\log \mathcal{L}_m$ has spread σ nats and is approximately Gaussian across members, then to leading order*

$$\mathbb{E}[\log \hat{\mathcal{L}}_M] - \log \mathcal{L} \simeq -\frac{1}{2} \frac{\text{Var}(\hat{\mathcal{L}}_M)}{\mathcal{L}^2} = -\frac{e^{\sigma^2} - 1}{2M}, \quad \frac{\text{ESS}}{M} \simeq e^{-\sigma^2}. \quad (24)$$

Requiring $|\text{bias}| < \epsilon$ nats therefore requires

$$M > \frac{e^{\sigma^2} - 1}{2\epsilon} \quad (25)$$

Proof. $\mathcal{L}_m$ lognormal with $\mathbb{E}[\mathcal{L}_m] = \mathcal{L}$ gives $\text{Var}(\mathcal{L}_m)/\mathcal{L}^2 = e^{\sigma^2} - 1$ and hence $\text{Var}(\hat{\mathcal{L}}_M)/\mathcal{L}^2 = (e^{\sigma^2} - 1)/M$. Expand log about $\mathcal{L}$ to second order. $\square$

(25) is sharper than the usual “ $M \sim e^{\sigma^2}$ ” folklore and it is brutal: at the plan’s upper estimate $\sigma \simeq 2.6$ nats, $e^{\sigma^2} = 863$, so $\epsilon = 0.1$ nat needs $M > 4.3 \times 10^3$, and $M = 8$ carries a 54-nat bias. That branch is excluded by measurement on the production line: σ is 1.49×10^{-2} nats at the anchor and at most 1.15 nats anywhere in the H_0 prior, and the measured θ -varying bias at $M = 8$ is 7.1×10^{-3} nats. The lognormal premise of (24) also fails at the low- H_0 rail, in the safe direction — the member log-weights there are left-skewed and heavy-tailed (skew -7.43 , excess kurtosis 74.1) and the measured $\text{ESS}/M = 0.835$ against a predicted $e^{-\sigma^2} = 0.265$ — so (25) is an upper bound on this line rather than an estimate.

Remark 4 (what actually has to be gated: the frozen-noise surrogate). The whitened draws g_m in (13) are generated *once* and reused at every θ (common random numbers). Therefore $\hat{\mathcal{L}}_M(\theta)$ is a **deterministic, smooth function of θ** , not a noisy one: the sampler explores an exactly reproducible surrogate surface. The bias of Proposition 3 is a fixed perturbation, and only its θ -variation

$$\Delta(\theta) = \log \hat{\mathcal{L}}_M(\theta) - \log \mathcal{L}(\theta), \quad \text{osc } \Delta \equiv \max_{\theta} \Delta - \min_{\theta} \Delta \quad (26)$$

distorts the posterior; a constant offset is absorbed into the evidence. The consequence for validation is unforgiving: every determinism diagnostic (repeat-to-repeat identity, adjacent- θ smoothness) passes *by construction* under CRN, and passes just as well on a badly distorted surrogate. Only (26), measured against a large- M reference, discriminates.

2.7 The $K \geq 2$ shared-latent joint solve

For K catalogs sharing one field, the stacked design is $\Phi'_k = b_k \Phi_k$ with $b = 1$ passed to the solver, so that the solver's default $\sigma_{\text{vox}}^2 = \sum_i (\Phi'_k)_{vi}^2 = b_k^2 \sum_i (\Phi_k)_{vi}^2$ is the per-survey-correct mean-one shift automatically. Then

$$J(\xi) = \frac{1}{2} \|\xi\|^2 + \sum_{k=1}^K \sum_{v \in \mathcal{V}_k} [\lambda_{kv} - N_{kv} \log \lambda_{kv}], \quad \lambda_{kv} = \bar{\lambda}_{kv} e^{b_k (\Phi_k \xi)_v - \frac{1}{2} b_k^2 \sigma_{kv}^2}, \quad (27)$$

$$H = \mathbb{K} + \sum_{k=1}^K b_k^2 \Phi_k^\top \text{diag}(\lambda_k) \Phi_k, \quad (28)$$

one Laplace ensemble $\{\xi_m\}$, and per-survey tables $\log Q_k^{(m)} = b_k \phi^\top \xi_m - \frac{1}{2} b_k^2 \sigma_{\text{prior}}^2$. Member m of every catalog is then the same physical realisation — a theorem, not a check — and it is stamped as a shared `realization_set_id` which the inference verifies.

Remark 5 (three verified limitations of this path, as of 0c5b3db). 1. **It is per-pixel-only.** The joint builder calls the shared assembly without a `c_mode` argument,

```
_assemble_gp3d_survey(path, cosmo=cosmo, survey=survey_k, z_s=z_s, edges_s=edges_s)
                        [build_joint_lognormal_completion.py:180-181]
```

so it silently takes the default `c_mode="per_pixel"` and the occupied-only base. Neither `aggregate` nor `selection` is reachable at $K \geq 2$.

2. **It never fixes the budget gauge.** `renormalize_q_mean_one` is called by the radial builder (:582) and the $K = 1$ gp3d builder (:990) and by *nothing* in `build_joint_lognormal_completion.py`. The $K \geq 2$ tables therefore carry the full Jensen monopole of Lemma 1 — the effect measured at +55% budget inflation — which the $K = 1$ tables have removed.
3. **Its rank cannot resolve any physical scale.** The joint inducing grid is hard-coded at $M_{\text{sph}} = 32$, $M_z = 6$, $z_{\text{node,hi}} = 3$ ($M = 192$), giving a ζ node spacing $\log(1 + 3)/5 = 0.277$. The *hard* radial resolution guard demands $\ell_\zeta \geq 0.277$, i.e. $L_{\text{smooth}} \geq 0.277 (1 + z_{\text{ref}}) d\chi/dz \simeq 1.34 \text{ Gpc}$ at $z_{\text{ref}} \simeq 0.24$. Any correlation length a DESI-like survey can actually support (50–190 Mpc, $\ell_\zeta = 0.010\text{--}0.039$) hard-fails the guard.

Together these say the shared-latent $K \geq 2$ path is not merely unplumbed for selection completeness: *it cannot today produce a resolved, gauge-fixed, selection-based shared field at all.*

2.8 Which half of the selection $\times K \geq 2$ seam is open

This is worth stating precisely, because the two halves have moved apart.

The inference half is OPEN. At master 0c5b3db, $K \geq 2$ mixtures under `c_mode="selection"` assemble and evaluate end-to-end: per-catalog magnitude fits with per-catalog θ routing, homogeneous Schechter mixtures with one shared `M_faint_offset`, a per-catalog Q -table θ firewall (cosmology checked globally, survey parameters checked against *catalog k 's own* table), and a two-catalog likelihood fixture pinning the identical-catalogs-collapse-to- $K=1$ identity and the no-cross-routing swap invariance (`tests/test_multitracer_selection.py`). There is exactly one refusal left on this side: *stratified* selection is single-catalog only, because the full-sky stratum map is attached to the shared data bundle while a $K \geq 2$ mixture builds each catalog's views from its

own bundle, so the map never reaches catalog k 's `EMCatalog` and its per-stratum empty-pixel budget would be silently missing (`inference/parameters.py:460--470`). That is a data-plumbing limitation, not a modelling one.

The builder half is CLOSED. Remark 5 is the statement. Operationally: the joint builder is the *only* producer of a shared `realization_set_id` (`save_lss_completion_hdf5` mints a fresh `uuid4` whenever one is not supplied, so K independently built single-catalog tables carry K different ids), and $K \geq 2$ `--lss_marginalize` requires equal, non-null ids unless `--allow_unverified_shared_lss_member` is passed — which downgrades the matched shared-field prior to an independent-fields product prior, i.e. throws away exactly the cross-tracer coupling the multitracer analysis exists to exploit. So a $K \geq 2$ shared-latent LSS marginalisation under a `selection` (or even `aggregate`) completeness is not refused; it is **not constructible**.

Consequence for the proposal. The seam does not need to be closed by extending the joint builder — passing `c_mode` through `build_joint_lognormal_completion.py:180`, calling `renormalize_q_mean_on` and unfreezing $(M_{\text{sph}}, M_z, z_{\text{node,hi}})$ would close it, and that is a genuine short-term option. But the field-level hierarchy of §3 closes it *by construction*: one ξ shared across K tracers makes “member m of every catalog is the same realisation” a theorem, retires `realization_set_id` and the whole provenance firewall, and removes the builder’s frozen rank, frozen θ and frozen completeness mode simultaneously. That is the strongest available argument that the field belongs inside the HBI rather than in an exported table.

3 The proposed model

3.1 The hierarchy

Promote ξ from a table to a latent variable of the inference:

$$\begin{aligned}
p(H_0, \Lambda, \xi, \{\theta_k\}, \{b_k\}, b_{\text{GW}}, \{f_k\} \mid d_{\text{GW}}, \{d_{\text{gal},k}\}) &\propto \pi(H_0, \Lambda) \pi(\{\theta_k\}) \pi(\{b_k\}) \pi(b_{\text{GW}}) \pi(\{f_k\}) \\
&\times \underbrace{\mathcal{N}(\xi; 0, \mathbb{K}_M)}_{\text{(F1)}} \times \prod_{k=1}^K \underbrace{p(\{m_i\}_k \mid \{z_i\}_k, \theta_k)}_{\text{(F2) magnitudes}} \times \prod_{k=1}^K \underbrace{p(\{N_{kv}\} \mid \xi, \theta_k, b_k, f_k, \Lambda)}_{\text{(F3) counts}} \\
&\times \underbrace{p(d_{\text{GW}} \mid \{d_{\text{gal},k}\}, \xi, b_{\text{GW}}, \Lambda, \{\theta_k\})}_{\text{(F4) GW}} \quad (29)
\end{aligned}$$

[new]
with $f_k = \{f_{k,p}\}_p$ the per-pixel areal completeness field of catalog k (data with a mask-model uncertainty, not a free function). The single structural change from §2 is that ξ now appears in (F1), (F3) and (F4), so every cosmological proposal changes the galaxy-field likelihood consistently. Factor (F2) is the existing magnitude channel; the chain rule $p(\{z\}, \{p\}, \{m\}) = p(\{z\}) p(\{p\} \mid \{z\}) p(\{m\} \mid \{z\}, \{p\})$ is what licenses writing (F2) and (F3) side by side.

3.2 (F3): the count likelihood

On solve voxels $v = (p, g)$:

$$N_{kv} \sim \text{Poisson} \left[\underbrace{f_{k,p} C_k(v; \theta_k) \bar{N}_k(v; \Lambda, \theta_k)}_{\bar{\lambda}_{kv}(\theta_k, \Lambda)} \exp \left(b_k s_v - \frac{1}{2} b_k^2 \sigma_v^2 \right) \right], \quad (30)$$

[new]

$$\bar{N}_k(v) = \Omega_{\text{pix}} \int_{z_g^-}^{z_g^+} \bar{n}(z; \Lambda, \theta_k) dz. \quad (31)$$

Two consequences must be stated explicitly because they are C -side changes:

- $f_{k,p}$ appears in (30) and in the consumed missing budget, $dN_{\text{miss}} = [1 - f_p C(z; \theta)] \bar{n} Q$. Putting it on one side only inverts the sign of the inferred incompleteness: a pixel with 63% areal masking is currently told 63% of its galaxies are a *density deficit* rather than a *selection deficit*, and that deficit then re-enters the GW likelihood as a claim about where hosts are not.
- Under `c_mode="per_pixel"` the base $\bar{\lambda}_{kv}$ is itself a plug-in estimate of the same counts N_{kv} ; (30) is then circular and must be refused. `aggregate` and `selection` bases are functions of θ , not of the counts, and are admissible.

3.3 (F4): the GW-host intensity on the same field

The GW host intensity is built from the *same* s with its own bias:

$$\Lambda_{\text{host}}(p, z; \Lambda, \theta, \xi) = \psi(z; \Lambda) \bar{n}(z; \Lambda, \theta) \exp \left[b_{\text{GW}} s(p, z) - \frac{1}{2} b_{\text{GW}}^2 \sigma^2(p, z) \right], \quad (32)$$

[new] with ψ the merger-rate weighting. Thinning (32) by the same $f_p C$ gives the catalogued/missing split, and the dark-siren redshift–sky prior becomes

$$p(z, p \mid \mathcal{C}, \xi, \Lambda, \theta) = \frac{\sum_{j \in \mathcal{C}} \delta_{p,p_j} \mathcal{K}_j(z) w_j(\Lambda) + [1 - f_p C(z; \theta)] \psi \bar{n} e^{b_{\text{GW}} s(p, z) - \frac{1}{2} b_{\text{GW}}^2 \sigma^2(p, z)}}{\mathcal{Z}(\Lambda, \theta, \xi)}. \quad (33)$$

[new] Structurally this is (6) with $Q_p(z)$ replaced by an explicit function of ξ and b_{GW} — which is why the seam in the code is one gather: $\log Q \rightarrow b_{\text{GW}} (\text{row_fac}[p] \cdot \phi_\zeta[n]) - \rho$.

3.4 Double conditioning: the exact factorisation

The catalog enters twice — as the empirical spike train in (33) and as the data of (30). This is the central correctness question of the whole proposal.

Theorem 1 (no double use). *Let $\mathcal{C}$ denote the catalogued galaxies and $\mathcal{U}$ the missing ones. Then, exactly,*

$$p(\mathcal{C}, d_{\text{GW}} \mid \xi, \Lambda, \theta) = \underbrace{p(\mathcal{C} \mid \xi, \theta, \Lambda)}_{(\text{F3})} \times \underbrace{p(d_{\text{GW}} \mid \mathcal{C}, \xi, \Lambda, \theta)}_{(\text{F4})}, \quad (34)$$

and the second factor may be evaluated with the host prior (33) **without any double use of the catalog**, provided the missing-galaxy intensity is written as a function of ξ alone and never of $\mathcal{C}$.

Proof. (34) is the chain rule and is exact. For the second claim, the GW factor requires (i) the catalogued hosts, which are $\mathcal{C}$ — a conditioning event, used as known positions, not re-modelled — and (ii) the intensity of $\mathcal{U}$. By the thinning independence (5), $p(\mathcal{U} \mid \mathcal{C}, \xi) = p(\mathcal{U} \mid \xi)$, so the intensity of $\mathcal{U}$ is $[1 - f_p C] \Lambda_{\text{gal}}(\xi)$ and involves $\mathcal{C}$ nowhere. The catalog’s information about $\mathcal{U}$ has been fully routed through (F3) into the posterior for ξ . $\square$

Remark 6 (what breaks it). Theorem 1 fails in exactly two ways, and both are live:

1. **Empirical-Bayes plug-in.** If the missing intensity is evaluated at a point estimate $\hat{\xi}(\mathcal{C})$ — which is precisely what the current Q table is — then $\mathcal{C}$ appears inside the GW factor. Multiplying that by (F3) uses the same counts twice. Today the code is *safe* only because it never multiplies by (F3): it is a conditional model with an empirical-Bayes prior, not a joint one. Adding the count likelihood without promoting ξ to a latent is the single most dangerous half-measure available.
2. **The monopole, twice.** Factor the counts exactly by shell totals $T_{kg} = \sum_p N_{kpg}$:

$$p(\{N_{kpg}\} | \xi, \theta) = \underbrace{p(\{T_{kg}\} | \xi, \theta)}_{\text{budget / monopole}} \times \underbrace{p(\{N_{kpg}\} | \{T_{kg}\}, \xi, \theta)}_{\text{placement}}. \quad (35)$$

The calibration that currently pins $(\log_{10} n_0, \delta)$ is a fit to the *same* counts over $z \in [0.05, 0.3]$, delivered to the run as `--fixed_parameter_values`. That prior is a surrogate for the first factor of (35). Keeping both the calibration prior *and* the unconditional Poisson count likelihood squares the monopole information. The admissible combinations are exactly two: (a) conditional (multinomial) count likelihood + external calibration prior, or (b) unconditional Poisson count likelihood + a genuinely independent prior on (n_0, δ) . Combination (a) is recommended, and it is what §3.5 makes exact.

3.5 The exact θ -cancellation criterion for the conditional channel

Write the shell-integrated intensity

$$\Lambda_{pg}(\theta, \xi) = \int_{z_g^-}^{z_g^+} f_p C(z; \theta) \bar{n}(z; \Lambda, \theta) e^{b s_p(z) - \frac{1}{2} b^2 \sigma_p^2(z)} dz, \quad \pi_{pg} = \frac{\Lambda_{pg}}{\sum_{p'} \Lambda_{p'g}}. \quad (36)$$

Define the within-shell normalised profile

$$\omega_g(z; \theta) = \frac{C(z; \theta) \bar{n}(z; \Lambda, \theta)}{\int_g C \bar{n} dz}, \quad \int_g \omega_g dz = 1, \quad (37)$$

so that $\Lambda_{pg} = a_g(\theta, \Lambda) f_p \int_g \omega_g(z; \theta) e^{b s_p(z) - \frac{1}{2} b^2 \sigma_p^2(z)} dz$ with a_g p -independent.

Proposition 4 (exact cancellation). π_{pg} is exactly independent of θ (and of Λ , and of $n_0, \delta, \theta_{\text{sel}}, H_0, \Omega_m, w_0, w_a$) if and only if the base is p -separable within the shell — $\bar{\lambda}_p(z) = u_p a(z)$, true whenever C is sky-uniform — **and** the field is shell-constant, $s_p(z) = s_{pg}$ for $z \in g$. Under those two conditions

$$\pi_{pg} = \frac{f_p e^{b s_{pg} - \frac{1}{2} b^2 \sigma_{pg}^2}}{\sum_{p'} f_{p'} e^{b s_{p'g} - \frac{1}{2} b^2 \sigma_{p'g}^2}}. \quad (38)$$

Otherwise the residual θ -dependence is exactly a within-shell covariance,

$$\frac{\partial \log \pi_{pg}}{\partial \theta} = \frac{\int_g \partial_\theta \omega_g e^{b s_p} dz}{\int_g \omega_g e^{b s_p} dz} - \sum_{p'} \pi_{p'g} \frac{\int_g \partial_\theta \omega_g e^{b s_{p'}} dz}{\int_g \omega_g e^{b s_{p'}} dz}, \quad (39)$$

which vanishes identically when $e^{b s_p(z)}$ is constant across the shell.

Proof. Immediate from (36): any p -independent factor divides out of the ratio, and shell-constancy of s_p pulls the exponential out of the z -integral so that the remaining $\int_g \omega_g dz = 1$ removes all θ -dependence. (39) is $\partial_\theta \log \Lambda_{pg} - \sum_{p'} \pi_{p'g} \partial_\theta \log \Lambda_{p'g}$ with a_g cancelled. $\square$

Remark 7. Proposition 4 converts the plan’s “freeze the shell-response operator W ” from a modelling choice into a measurable residual: freezing W at θ_{ref} is exactly the approximation $\partial_\theta \omega_g \rightarrow 0$, and (39) is the quantity to bound. It also identifies the correct shell width: shells narrow enough that s is constant across them make the cancellation exact, but the radial resolution is bounded below by the photo- z scatter ($\sigma_z \simeq 0.023$, $\sigma_\chi \simeq 89 \text{ Mpc}$), so the honest statement is that the cancellation is exact only up to the photo- z -smoothed radial structure — which is precisely the structure the forward convolution puts into W .

3.6 The per-proposal conditional solve and its warm start

For each proposal $(\Lambda, \theta, \{b_k\}, b_{\text{GW}})$ define

$$J_\theta(\xi) = \frac{1}{2} \|\xi\|^2 - \log p(\{N\} \mid \xi, \theta), \quad (40)$$

with, in the Poisson form, ∇J_θ and $H(\theta)$ as in (12), and in the conditional multinomial form

$$\nabla J_\theta = \xi - b \sum_g \Phi_g^\top (N_g - T_g \pi_g), \quad (41)$$

$$H(\theta) = \mathbb{K} + b^2 \sum_g T_g \left[\Phi_g^\top \text{diag}(\pi_g) \Phi_g - u_g u_g^\top \right], \quad u_g = \Phi_g^\top \pi_g. \quad (42)$$

[new]

Derivation of (42). With $\eta_p = b s_p$ and $\pi_p = e^{\eta_p} / \sum_{p'} e^{\eta_{p'}}$, the multinomial log-likelihood $\ell = \sum_p N_p \log \pi_p$ has $\partial \ell / \partial \eta_p = N_p - T \pi_p$ and $\partial^2 \ell / \partial \eta_p \partial \eta_q = -T(\delta_{pq} \pi_p - \pi_p \pi_q)$. Chain through $\eta = b \Phi \xi$ and negate. $\square$

The rank-one subtraction $-T_g u_g u_g^\top$ is not optional: it is a positive-semidefinite term, so omitting it makes H too large, H^{-1} too small, and the Laplace draws *under*-dispersed — exactly the direction that hides the error of Proposition 3. It is separable and therefore free: with $\Phi_g = \Phi_{\text{sph}} \otimes \phi_\zeta[g]$,

$$u_g = v_g \otimes \phi_\zeta[g], \quad v_g = \Phi_{\text{sph}}^\top \pi_g, \quad T_g u_g u_g^\top = T_g (v_g v_g^\top) \otimes (\phi_\zeta[g] \phi_\zeta[g]^\top). \quad (43)$$

Proposition 5 (warm starts are admissible under a gradient stopping rule). *Let $\xi_\star$ be the iterate at which the solve stops, with $\varepsilon = \|\nabla J_\theta(\xi_\star)\|_2$. Since $H \succeq \mathbb{K}$ (the whitened prior),*

$$0 \leq J_\theta(\xi_\star) - J_\theta(\hat{\xi}_\theta) \simeq \frac{1}{2} \nabla J^\top H^{-1} \nabla J \leq \frac{\varepsilon^2}{2 \lambda_{\min}(H)} \leq \frac{\varepsilon^2}{2}. \quad (44)$$

Hence any start — including a warm start from a neighbouring proposal — yields a likelihood that is history-independent to within $\varepsilon^2/2$ nats.

Remark 8. The blanket prohibition on warm starts is too strong and the blanket permission is too weak. (44) gives the operational rule: stop on $\|\nabla J\|_2 < \varepsilon$ with $\varepsilon^2/2$ at least an order of magnitude below the θ -oscillation budget of (26). For a 0.01-nat budget, $\varepsilon < 0.14$. What is genuinely inadmissible is a *fixed iteration count* started from a history-dependent point, because then the residual is unbounded and $\log \mathcal{L}$ becomes a functional of the sampler’s traversal order — at which point nested sampling’s $\mathbb{E}[\log X_i] = -i/n$ shrinkage argument no longer refers to any well-defined likelihood. A deterministic anchored start $\xi^{(0)} = \hat{\xi}(\theta_{\text{ref}})$ satisfies both requirements trivially and costs one or two extra Newton steps.

3.7 Laplace marginalisation of the field

Proposition 6 (the galaxy-side evidence). *With the whitened prior $\mathcal{N}(\xi; 0, \mathbb{K}_M)$,*

$$\log \mathcal{L}_{\text{gal}}(\theta) = \log \int d\xi \mathcal{N}(\xi; 0, \mathbb{K}_M) p(\{N\}|\xi, \theta) \simeq -J_\theta(\hat{\xi}_\theta) - \frac{1}{2} \log \det H(\theta), \quad (45)$$

with **no** residual $(2\pi)^{\pm M/2}$: the prior normalisation and the Gaussian integral cancel exactly. Numerically $\log \det H = 2 \sum_i \log(L_H)_{ii}$.

Proof. $\int (2\pi)^{-M/2} e^{-J(\xi)} d\xi \simeq (2\pi)^{-M/2} e^{-J(\hat{\xi})} (2\pi)^{M/2} |H|^{-1/2}$. □

The θ -gradient, for gradient-based samplers, is

$$\frac{\partial}{\partial \theta} \log \mathcal{L}_{\text{gal}} = - \left. \frac{\partial J_\theta}{\partial \theta} \right|_{\hat{\xi}_\theta} - \frac{1}{2} \text{tr} \left(H^{-1} \frac{dH}{d\theta} \right), \quad \frac{dH}{d\theta} = \frac{\partial H}{\partial \theta} + \sum_i \frac{\partial H}{\partial \xi_i} \frac{\partial \hat{\xi}_i}{\partial \theta}, \quad (46)$$

where the envelope theorem removes $\partial \hat{\xi}/\partial \theta$ from the first term (because $\nabla_\xi J = 0$ at $\hat{\xi}$) but *not* from the second, and the implicit function theorem gives $\partial \hat{\xi}/\partial \theta = -H^{-1} \partial_\theta \nabla_\xi J$.

Exactness. (45) is exact iff J is quadratic in ξ . The controlling small parameter is the posterior standard deviation of the linear predictor s_v , measured at $\simeq 5.6 \times 10^{-3}$; the leading Edgeworth correction to the log-evidence is $O(10^{-3})$ nats. This is *not* the estimator error of Proposition 3, which is larger by orders of magnitude and is a separate budget.

3.8 The fixed-whitened-draw alternative and its ESS law

The cheap alternative to a per-proposal solve is to draw $\{g_m\}$ once and reuse the resulting $\{\xi_m\}$ at every θ — exactly what the member ensemble does today. Formally this is importance sampling with proposal $q = \mathcal{N}(\hat{\xi}_0, H_0^{-1})$ and target $p(\xi|\{N\}, \theta) \simeq \mathcal{N}(\hat{\xi}_\theta, H_\theta^{-1})$:

$$\hat{\mathcal{L}}(\theta) = \frac{\sum_m w_m(\theta) \mathcal{L}_{\text{GW}}(\theta; \xi_m)}{\sum_m w_m(\theta)}, \quad w_m(\theta) = \frac{\mathcal{N}(\xi_m; 0, \mathbb{K}) p(\{N\}|\xi_m, \theta)}{q(\xi_m)}. \quad (47)$$

Proposition 7 (the ESS law for a fixed draw set). *If $H_\theta \simeq H_0 \equiv H$ and only the mode moves, then under $\xi \sim q$*

$$\log w(\theta) = (\hat{\xi}_\theta - \hat{\xi}_0)^\top H (\xi - \hat{\xi}_0) + \text{const}, \quad \text{Var}[\log w] = \|\hat{\xi}_\theta - \hat{\xi}_0\|_H^2, \quad (48)$$

and hence

$$\boxed{\frac{\text{ESS}(\theta)}{M_{\text{draw}}} \simeq \exp\left(-\|\hat{\xi}_\theta - \hat{\xi}_0\|_H^2\right)} \quad (49)$$

where $\|x\|_H^2 = x^\top H x$ is measured in units of the posterior standard deviation, summed over all M modes.

Proof. For two Gaussians with common precision H , the log ratio is affine in ξ with coefficient $H(\hat{\xi}_\theta - \hat{\xi}_0)$; the variance of an affine form under q is $(\hat{\xi}_\theta - \hat{\xi}_0)^\top H H^{-1} H (\hat{\xi}_\theta - \hat{\xi}_0)$. Then $\text{ESS}/M = 1/(1 + \text{Var}[w]/\mathbb{E}[w]^2) = e^{-\text{Var}[\log w]}$ for lognormal w . □

Remark 9 (the criterion is on the total, not the per-mode, displacement). (49) is the exact criterion for when the compressed ensemble is valid, and it is **much** stricter than a per-mode statement. A tolerance written as $\|\Delta\hat{\xi}\|_H/\sqrt{M} < \tau$ implies $\|\Delta\hat{\xi}\|_H^2 < \tau^2 M$ and therefore $\text{ESS}/M \simeq e^{-\tau^2 M}$. At the plan's rank $M = 3780$ and its proposed $\tau = 0.1$, this is $e^{-37.8} \approx 4 \times 10^{-17}$ — a complete collapse, from a pin that reads as comfortably tight. To keep $\text{ESS}/M > 0.1$ one needs

$$\|\hat{\xi}_\theta - \hat{\xi}_0\|_H^2 < \log 10 \simeq 2.3 \iff \tau < \sqrt{2.3/M} = 0.025 \text{ at } M = 3780. \quad (50)$$

The same measured number $\|\hat{\xi}_\theta - \hat{\xi}_0\|_H$ therefore plays two distinct roles and needs two distinct tolerances:

- (a) as a *model-misspecification* diagnostic — how far the frozen model's field sits from the exact θ -coupled one — a per-mode tolerance $\|\Delta\hat{\xi}\|_H/\sqrt{M} < 0.1$ is a defensible scale, because what matters there is whether the fitted field looks different;
- (b) as the *estimator-validity* criterion for reusing one draw set across θ , only the total (50) is admissible, because importance weights compound over all M modes.

Any validation of the fixed-draw path must be stated at (50), not at a per-mode fraction. This is the quantitative reason the θ -freezing of the conditional channel (Proposition 4) is worth its cost: when the cancellation is exact there is no importance sampling at all — it sets $\hat{\xi}_\theta \equiv \hat{\xi}_0$ identically and makes the ESS perfect by construction rather than by measurement, and only role (a) remains.

Proposition 8 (when the compressed ensemble is *exact*). *If $p(\xi | d_{\text{gal}}, \theta) = p(\xi | d_{\text{gal}})$ for all θ , then a single draw set $\{\xi_m\} \sim p(\xi | d_{\text{gal}})$ gives, at every θ , an unbiased estimator of $\mathcal{L}(\theta)$ with unit importance weights. The only error is the Monte-Carlo error of Proposition 3; there is no importance-sampling degeneracy at all. Conversely, the full θ -coupled solve is **required** whenever*

$$\|\hat{\xi}_\theta - \hat{\xi}_0\|_H^2 \gtrsim \log\left(\frac{M_{\text{draw}}}{\text{ESS}_{\min}}\right). \quad (51)$$

4 The three limits: one hierarchy, three known analyses

Write the single-event likelihood as

$$\mathcal{L}_i(\Lambda, \theta, \xi) = \int dz d\hat{n} \, p_i(d_i | d_L(z; \Lambda), \hat{n}) \frac{\Lambda_{\text{cat}}(\hat{n}, z) + \Lambda_{\text{miss}}(\hat{n}, z; \xi)}{\int [\Lambda_{\text{cat}} + \Lambda_{\text{miss}}]}. \quad (52)$$

All three established analyses are limits of (52), and the field marginalisation is exactly the operation that interpolates between them.

4.1 Limit I — complete catalog, good localisation: the discrete host sum

$f_p C \rightarrow 1$ implies $\Lambda_{\text{miss}} \rightarrow 0$ and Λ_{cat} is the empirical point process. Then

$$\mathcal{L}_i(\Lambda) \longrightarrow \sum_{j \in \Omega_{90}^{(i)}} w_j(\Lambda) p_i(d_i | d_L(z_j; \Lambda), \hat{n}_j), \quad (53)$$

the classical statistical standard siren. ξ drops out of (52) identically, so $p(\xi | d_{\text{gal}}, d_{\text{GW}}) = p(\xi | d_{\text{gal}})$ *exactly* and the marginalisation is trivial. **Exact.**

4.2 Limit II — incomplete, good localisation: spikes plus a clustered missing field

Λ_{tot} is a sum of catalog spikes and a smooth clustered component. Because a well-localised event's support spans only $O(1-10)$ correlation cells at the smoothing scale L_{smooth} , the likelihood depends on ξ essentially through the *point value* $s(\hat{n}_i, z)$ in the localisation volume. Expanding to second order in the field,

$$\log \mathbb{E}_\xi[\mathcal{L}_i] \simeq \log \bar{\mathcal{L}}_i + b_{\text{GW}} \langle s \rangle_i + \frac{1}{2} b_{\text{GW}}^2 [\text{Var}_\xi \langle s \rangle_i - \langle \sigma^2 \rangle_i] + O(b^3), \quad (54)$$

where $\langle \cdot \rangle_i$ is the average against the event's normalised localisation kernel restricted to the missing component. The first-order term is the placement information; the second-order term is the placement *uncertainty*, and it is exactly what the member ensemble estimates. **Exact given the model; the expansion (54) is $O(b^3)$.**

4.3 Limit III — very incomplete, diffuse localisation: a correlation-function statement

Now $\Lambda_{\text{cat}} \rightarrow 0$ and the localisation spans many correlation cells. Write $p_i(x)$ for the event's normalised localisation density on $x = (\hat{n}, z)$. Expanding (52) in b_{GW} s and taking the field expectation over $\xi \sim \mathcal{N}(\hat{\xi}, \Sigma_\xi)$,

$$\log \mathbb{E}_\xi \left[\prod_i \mathcal{L}_i \right] \ni b_{\text{GW}}^2 \sum_{i < j} \iint p_i(x) p_j(y) \xi_{\text{mm}}(|x - y|) d^3x d^3y + b_{\text{GW}} b_{\text{gal}} \sum_i \iint p_i(x) \delta_g(y) \xi_{\text{mm}}, \quad (55)$$

i.e. **the field-marginalised dark-siren likelihood depends on the field only through the two-point function smeared by the events' localisation kernels.** Decomposing the localisation kernels in spherical harmonics per tomographic shell, $p_i(\hat{n}, z) = \sum_{\ell m} p_{\ell m}^i(z) Y_{\ell m}(\hat{n})$, (55) becomes

$$b_{\text{GW}}^2 \sum_\ell \frac{2\ell + 1}{4\pi} \mathcal{W}_\ell^i \mathcal{W}_\ell^j C_\ell^{gg}(z_i, z_j) + b_{\text{GW}} b_{\text{gal}} \sum_\ell \frac{2\ell + 1}{4\pi} \mathcal{W}_\ell^i \mathcal{W}_\ell^g C_\ell^{\text{GW} \times g}, \quad (56)$$

which is the harmonic cross-correlation estimator, with the GW localisation windows $\mathcal{W}_\ell^i$ playing the role of tomographic bin windows. The bias combination that appears, $b_{\text{GW}} b_{\text{gal}}$, is exactly the combination a $C_\ell^{\text{GW} \times g}$ measurement constrains. **Approximate: (55)–(56) are the $O(b^2)$ truncation of (52), and are exact only in the Gaussian, weakly-clustered regime.**

4.4 The unification statement

Limits I–III are not three methods; they are one likelihood evaluated at three completeness/localisation regimes. Catalog dark sirens are the $\Lambda_{\text{miss}} \rightarrow 0$ corner; cross-correlation is the $\Lambda_{\text{cat}} \rightarrow 0$, many-cell corner; the field-level hierarchy of §3 is the interpolant that is valid at both ends and, unlike either, in the middle — which is where every real catalog/detector pairing actually sits. The operational content of the unification is Proposition 8: whether a given analysis *needs* the in-likelihood field is decided by (51), a number one can measure, not by which corner one prefers.

5 The per-proposal algorithm

5.1 Offline (once per catalog set)

```

ANCHOR BUILD -- inputs: catalogs {k}, masks {f_k}, anchor (theta_ref, Lambda_ref)
A1 basis:    Zn, Zz <- inducing nodes (M_sph x M_z); L <- chol(K_ZZ + j*I)
              Phi_sph (N_pix x M_sph),  Phi_zeta (N_z x M_z)      [never dense Phi]
A2 counts:   N_kpg <- histogram(z_gal, bins=z_count_edges)      [integer, exact]
A3 shell response W (G_s x N_fine):
              W[g,n] = (int_g K(z|z_n) dz) * omega_g(z_n; theta_ref) * Delta_n
              rows normalized to 1      [photo-z convolved FORWARD, not on data]
A4 solve:    xi_hat, H <- Newton on J_{theta_ref} [Poisson or multinomial
              grad/Hessian, Sec. 3.6]
A5 draws:    xi_m = xi_hat + L_H^{-T} g_m, g_m antithetic pairs, seed stamped
A6 moments:  A_m(z;b) = sum_p e^{b f_m(p,z)}, B_m(z;b) = sum_p f_p e^{b f_m(p,z)}
              on a Chebyshev grid in b; P = sum_p 1, F = sum_p f_p
A7 stamp:    sha256 over (M_sph, M_z, ls_ang, ls_zeta, amp, jitter_mode, j, nside,
              hash(f_p), hash(W), z_count_edges, hash(counts), theta_ref, {b_k})

```

Step A6 is what removes the full-sky reduction from the online path. The budget normaliser with the consumption weights is

$$\rho_m(z; \theta, b) = \log \frac{\sum_p [1 - f_p C(z; \theta)] e^{b f_m(p, z)}}{\sum_p [1 - f_p C(z; \theta)]} = \log \frac{A_m(z; b) - c B_m(z; b)}{P - c F}, \quad c \equiv C(z; \theta), \quad (57)$$

so the only online θ -dependence is through the *scalar* $c \in [0, 1]$ at each z . A_m, B_m are θ -free by construction. Equation (57) is exact, not an interpolation, given A_m, B_m on the b -grid.

5.2 Online (once per proposal)

LIKELIHOOD(H0, Lambda, {theta_k}, {b_k}, b_GW):

```

-- decide the path -----
P0 if base is p-separable within every shell      (c_mode in {aggregate,
      AND the frozen-W residual is below budget      selection})
      [the exact-cancellation criterion, Prop. "exact cancellation"]
      then FAST PATH      (theta-cancellation holds; xi posterior is theta-free)
      else FULL PATH

-- FAST PATH: compressed ensemble, exact by Prop. "compressed exactness" --
F1 c(z) <- C(z; theta)      [scalar per z node]
F2 rho_m <- (A_m - c B_m) / (P - c F)      closed form, budget normalizer
F3 for each member m:
      logQ_m(p,z) = b_GW * (row_fac_m[p] . phi_zeta[z]) - log rho_m(z)
      ll_m = PE_terms(logQ_m) - N_ev * log mu(logQ_m)
F4 logL = LSE_m(ll_m) - log M_draw
F5 diagnostics: member ESS = exp(-sum_m p_m log p_m), p_m = softmax(ll_m)

-- FULL PATH: theta-coupled conditional solve -----

```



```

G1  xi <- xi_hat(theta_ref)                                [DETERMINISTIC anchor, not the previous
                                                            proposal's solution unless G3 holds]
G2  Newton/Fisher-scoring on J_theta:
      grad = xi - b sum_g Phi_g^T (N_g - T_g pi_g)
      H     = I + b^2 sum_g T_g [ Phi_g^T diag(pi_g) Phi_g - u_g u_g^T ]
      (separable two-stage contraction; never form S[p,(i,j)])
G3  stop when ||grad||_2 < eps with eps^2/2 << osc-budget
      [Prop. "warm starts are admissible": J(xi*) - J(xi_hat) <= eps^2/2]
G4  logL_gal = -J_theta(xi_hat) - 0.5*logdet H, logdet H = 2 sum log diag(L_H)
G5  draws: xi_m = xi_hat + L_H^{-T} g_m                    [g_m FIXED across theta: CRN]
G6  logL = logL_gal + LSE_m(ll_m(xi_m)) - log M_draw

-- the exactness switch -----
S1  monitor d2 = || xi_hat(theta) - xi_hat(theta_ref) ||_H^2
S2  if d2 > log(M_draw / ESS_min) -> FAST PATH is invalid, use FULL PATH
      [the ESS law; note d2 is a SUM over all M modes, NOT a per-mode average,
       so a "0.1 posterior sd per mode" pin means d2 = 0.01*M and fails]

```

5.3 Cost structure of the two paths

The fast path adds work only to the member-vmapped slice: the seam replaces a gather `member_logq[pix,idx]` by a contraction `row_fac[pix] @ phi_zeta[idx]`. The row factors are $M_{\text{draw}} \times n_{\text{rows}} \times M_z$ instead of $M_{\text{draw}} \times n_{\text{rows}} \times N_{\text{grid}}$ — a compression factor $N_{\text{grid}}/M_z \simeq 90$, which is what makes a defensible M_{draw} affordable at all under (25). The full path adds, per proposal, one $M \times M$ Cholesky ($M^3/3$) plus the separable contraction $2G_s N_{\text{pix}} M_{\text{sph}}^2 + 2M_{\text{sph}}^2 G_s M_z^2$; at $(M_{\text{sph}}, M_z, G_s) = (315, 12, 32)$ this is $O(1)$ s of dense linear algebra per proposal. The 27.5–49.3 ms likelihood baseline this was first compared against belongs to a different configuration: the production 259-event line is *measured* at 1417 ms per call with no structure (H100, value-only path), so the full path is of order the likelihood itself rather than four orders above it. **The full path is still not affordable inside a 10^6 -call nested-sampling run at production rank**, but the binding costs are memory — it makes the member row factors a 3–24 GB per-evaluation transient — and the loss of an analytically smooth surrogate, not wall clock; it is affordable at reduced rank, on a grid, or as a validation reference. This asymmetry — not elegance — is why Proposition 4 is the load-bearing result.

6 Exact / approximate ledger

statement	status	controlling parameter
Thinning independence (5)	exact	—
Chain-rule factorisation (34)	exact	—
$\mathbb{E}[Q]$ variance shift, Lemma 1	exact	given the Laplace posterior
Monopole gauge, Prop. 1	exact	—
Budget identity in <code>aggregate/selection</code> , Prop. 2(1)	exact	sky-uniform C
Shell-total factorisation (35)	exact	—
θ -cancellation (38)	exact iff	p -separable base <i>and</i> shell-constant s
ESS law (49)	exact for	Gaussian mode shift at fixed H
Limit I (53)	exact	$f_p C \rightarrow 1$
Low-rank Nyström $\Phi\Phi^\top \approx K$	approx.	rank M vs. $\ell_{\text{ang}}, \ell_\zeta$; hard guard
Laplace evidence (45)	approx.	posterior sd of s (5.6×10^{-3}) $\Rightarrow$ $O(10^{-3})$ nats
Member estimator (22)	approx.	$(e^{\sigma^2} - 1)/2M$, eq. (25)
Deterministic- Q plug-in, Rem. 1	approx.	first order in posterior spread
Renormalise-then-average, Rem. 3	approx.	$\text{Var}_{\text{sky}}(Q)/N_{\text{pix}}$
Budget identity in <code>per_pixel</code> , Prop. 2(2)	approx.	footprint fraction; $\theta - \theta_{\text{ref}}$
Frozen W	approx.	residual (39)
Population-average photo- z kernel in W	approx.	spread of σ_z across the sample
Warm start	approx.	$\varepsilon^2/2$, eq. (44)
Limit II (54)	approx.	$O(b_{\text{GW}}^3)$
Limit III (56)	approx.	$O(b^2)$ truncation; Gaussianity
$Q \equiv 1$ above z_{depth} and off-footprint	<i>choice</i>	zero assumed variance where there is no data

Table 2: What is exact and what is not. The three “choice” rows are modelling conventions and must be quoted as systematics, never as marginalisations.